

The Pricing of Biodiversity Risk in Commodity Markets

Replication and USDA AMS Extension

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Advanced Methods for Portfolio and Risk Management

May 17, 2026

- 1 Introduction
- 2 Original Paper and Replication
- 3 USDA AMS Extension
- 4 Robustness
- 5 Conclusion

Introduction

Biodiversity loss is emerging as a material risk factor for financial markets, alongside the more familiar climate-change channel.

The paper by Guidolin and Pedio (*Review of Finance*, 2026) asks a sharp question:

Central question

Do commodity markets price biodiversity-related **transition risk**? Do commodities with higher biodiversity exposure command a risk premium, and do they react differently to binding policy news?

The mechanism operates through two channels.

- **Normal times:** higher biodiversity intensity $\Rightarrow$ greater regulatory and reputational exposure $\Rightarrow$ higher required excess return (risk premium).
- **Around the Kunming Declaration (Oct 2021):** the event raises the salience of biodiversity transition risk, triggering a *negative* price revision for high-intensity commodities — to allow for larger future returns.

Project Structure

This project has two goals.

- 1 **Replication.** Reproduce the central empirical results using the original pipeline: IMF spot prices, JNCC/FAO biodiversity footprint data, and commodity-level controls. The focus is on (i) the baseline biodiversity risk premium and (ii) the Kunming event study.
- 2 **Extension.** Provide external validation using an independent data source: USDA AMS local cash prices (MyMarketNews/MARS). The panel is observed at the *commodity* $\times$ *local U.S. market* $\times$ *month* level. The goal is to test whether the Kunming repricing also appears in disaggregated local cash markets — not only in international spot prices.

The Original Paper: Measuring Biodiversity Exposure

Why intensity, not total species loss?

Total predicted species loss is strongly driven by production scale: Russia and Australia rank highest for barley *loss* mainly because they produce more, not because production is more destructive.

The paper therefore uses an **intensity** measure:

$$\text{Int. Species Loss}_{i,t} = \frac{\text{Predicted species loss}_{i,t}}{\text{Crop area}_{i,t}}$$

This captures biodiversity damage *per unit of land* and abstracts from scale effects. It is the relevant exposure for transition risk.

Key event: Kunming Declaration

On October 13, 2021, countries committed to reducing biodiversity loss, including land- and sea-use change, and promoting sustainable production systems by 2030.

Two relevant targets:

- **Target 1:** bring ecosystem degradation close to zero.
- **Target 16:** reduce food consumption's biodiversity footprint.

Prediction: more exposed commodities should experience an immediate price decline after the declaration, implying higher expected future returns.

Replication Data and Descriptive Statistics

Dataset: 23 agricultural commodity series, monthly spot returns, 2005–2022 (IMF); biodiversity data from the JNCC/FAO Commodity Footprints project.

Table: **Summary statistics**

	Mean	SD	p25	p50	p75
<i>Panel A: Biodiversity exposure (annual)</i>					
Intensity species loss	9.76	13.48	2.10	4.26	14.30
Deforestation, 1,000 ha (log)	3.33	1.62	2.17	3.49	4.44
CO ₂ emissions, 10,000 t (log)	2.23	1.82	1.04	2.40	3.57
<i>Panel B: Commodity returns (monthly, %)</i>					
Excess return (spot)	0.50	7.61	−3.17	0.17	3.92
Excess return (future)	0.42	8.43	−4.69	0.06	5.27

The wide dispersion in intensity ($SD > \text{mean}$) reflects the large heterogeneity across commodities: coffee and cocoa reach values above 15 species per 10,000 km²; barley and wheat remain below 2.5. The paper validates this ranking against the ecology literature — all top-8 commodities are produced in tropical biodiversity hotspots.

Baseline Pricing Regressions

Baseline specification with commodity and month–year fixed effects:

$$r_{i,m,t} = \beta_1 \text{IntSpeciesLoss}_{i,t-1} + \beta_2 \text{Controls}_{i,t-1} + \mu_m + \gamma_i + \varepsilon_{i,m,t}.$$

Table: Baseline panel regressions

	Panel A: Excess return			Panel B: Risk-adjusted return		
	(1)	(2)	(3)	(4)	(5)	(6)
Int. sp. loss _{<i>t</i>−1}	0.053*** (0.011)	0.059*** (0.014)	0.061*** (0.015)	0.052*** (0.015)	0.056*** (0.018)	0.060*** (0.021)
Prod. growth _{<i>t</i>−1}		−0.011 (0.022)	−0.011 (0.022)		−0.007 (0.021)	−0.006 (0.021)
Log CO ₂ _{<i>t</i>−1}		−0.048 (0.341)			0.012 (0.362)	
Log defor. _{<i>t</i>−1}			0.060 (0.405)			0.133 (0.425)
Obs.	4,692	4,344	4,416	4,692	4,344	4,416
FE	Commodity + month–year					

SE double-clustered by month–year and commodity. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Biodiversity intensity is positively and significantly priced. A 10-unit increase implies about 58 bp/month higher spot returns; CO₂ and deforestation are insignificant.

Sub-Sample Evidence

The premium appears to be a *recent* phenomenon. Splitting the sample at January 2015, around the release of *Global Biodiversity Outlook 4*, shows:

	Full sample	Before 2015	2015–2022
Int. species loss _{<i>t</i>−1} , raw ret.	0.053*** (0.011)	−0.026 (0.089)	0.135*** (0.022)
Int. species loss _{<i>t</i>−1} , risk-adj.	0.052*** (0.015)	−0.065 (0.080)	0.125*** (0.025)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Before 2015 the coefficient is small and insignificant; after 2015 it becomes large and highly significant. This supports the event-study logic: markets started internalizing biodiversity transition risk as global regulatory discourse intensified.

Kunming Event Study

To test whether high-intensity commodities were repriced around the Kunming Declaration, we estimate:

$$r_{i,m,t} = \delta_1(\text{Post}_t \times \text{HighIntensity}_i) + \mu_m + \gamma_i + \varepsilon_{i,m,t}.$$

$\text{Post}_t = 1$ only in **November 2021**; $\text{HighIntensity}_i = 1$ if intensity is above the 2020 median.

Table: Kunming event study

Outcome	Specification	Coef.	SE
Excess return	Static: Post only	−3.587***	0.213
Abnormal return	Static: Post only	−2.729***	0.236
Excess return	Dynamic: $m-1$	+1.100	1.332
Excess return	Dynamic: $m=0$ (Oct.)	−2.931	1.821
Excess return	Dynamic: $m+1$ (Nov.)	−3.595***	1.124
Excess return	Dynamic: $m+2$	+0.136	1.139

Commodity and month–year FE. SE double-clustered by month–year and commodity. *** $p < 0.01$.

High-intensity commodities lose about 3.6 pp in November 2021 relative to low-intensity commodities. The effect is concentrated at $m+1$, with no significant pre-trend at $m-1$ and no persistence at $m+2$.

Dynamic Event Study — Coefficients

The dynamic specification over a $[-4, +4]$ window, with $m-1$ as baseline, gives the following point estimates:

Event month	Coefficient	SE
$m-4$	-0.448	0.663
$m-3$	+0.766	0.712
$m-2$	+3.222	0.703
$m=0$ (Oct. 2021)	-2.915***	0.922
$m+1$ (Nov. 2021)	-3.579***	0.483
$m+2$	+0.152	0.390
$m+3$	+1.443**	0.497
$m+4$	-0.483	0.529

Baseline period: $m-1$. ** $p < 0.05$; *** $p < 0.01$.

Pre-event coefficients are not systematically negative, suggesting no clear pre-existing differential trend. The repricing is concentrated at $m = 0$ and $m + 1$, then dissipates.

Note: point estimates are virtually identical to Figure 6 of the paper. Confidence intervals differ slightly because the Python/statsmodels replication implements the two-way clustered covariance matrix with small-sample adjustments that may not exactly match the original Stata output.

Placebo Test (Paris Agreement)

To confirm that intensity of species loss captures *biodiversity* transition risk and not climate risk, the same event study is run using the Paris Agreement (December 2015) as the focal date, keeping the same treated commodities.

Specification	Coef.	SE
Post $\times$ Treated (excess ret.)	+0.275	0.200
Post $\times$ Treated (abn. ret.)	+0.299	0.205
Dynamic $m+1$ (excess ret.)	-1.147	1.540
Dynamic $m+2$ (excess ret.)	+0.920	1.204

- None of the coefficients is statistically significant. High-intensity commodities did *not* react to a major climate policy event.
- Biodiversity and climate transition risks are two related but **distinct** risk factors. The intensity of species loss measure captures something that CO₂ exposure alone does not — which is precisely why it is informative for pricing.

Aggregate Biodiversity Risk: Portfolio Evidence

Beyond commodity-level regressions, the paper asks whether *aggregate* biodiversity risk is priced in commodity returns.

Method: estimate each commodity's beta to the Biodiversity Loss Index (BLI), sort commodities into tercile portfolios by $\hat{\beta}_{i, \text{BLI}}$, and rebalance annually. High-beta commodities suffer when aggregate biodiversity declines, so they should earn a premium.

Table: Portfolios sorted by biodiversity beta

Portfolio	Pre-form. $\hat{\beta}$	Mean ret. (%)	Std. dev. (%)	Alpha (%)
Low (hedge)	-1.439	0.177 (0.275)	3.274	0.042 (0.190)
Mid	-0.043	0.404 (0.289)	3.266	0.279 (0.188)
High (risk)	+1.554	0.539** (0.255)	3.187	0.458** (0.179)
High — Low		0.361* (0.192)	2.969	0.416** (0.204)

Alpha from regression on 7 commodity factors. Newey–West SE. * $p < 0.10$; ** $p < 0.05$.

The long-short alpha is 0.42%/month, about 5.3% annualized. The premium is monotone across terciles and survives standard commodity factor controls.

Why local cash prices?

- The original paper uses *international* IMF spot prices — aggregated, exchange-traded series.
- If biodiversity repricing is a genuine economic force it should propagate to *local* cash markets where physical delivery occurs and where financialization is minimal.
- A finding in USDA AMS prices constitutes **external validation** using a fully independent data source with richer cross-sectional variation (commodity $\times$ local market).

Data source

USDA AMS MyMarketNews / MARS

Daily local cash quotes at the level of commodity, local market, trade location, and delivery point. Aggregated to a monthly *commodity $\times$ local market* panel.

The same JNCC/FAO biodiversity intensity data are merged in at the commodity level, maintaining exact comparability with the replication.

USDA Data: Coverage

Table: USDA AMS data coverage

Sample	Raw quote obs.	Monthly obs.	Commodities	Comm.–market units
Main grain	713,928	22,095	6	433
Expanded add.	49,136	2,174	7	60
Full USDA	763,064	24,269	13	493

Table: Biodiversity intensity mapping — USDA commodities

Commodity	Int. loss	High?	Commodity	Int. loss	High?
Soybeans	1.121	No	Canola	1.881	Yes
Flaxseed	1.410	No	Wheat	2.198	Yes
Peas	1.456	No	Barley	2.385	Yes
Sunflower	1.595	No	Lentils	3.028	Yes
Oats	1.604	No	Beans	3.103	Yes
Millet	1.737	No	Corn	3.916	Yes
Sorghum	1.824	No			

High intensity = above-median within the USDA sample, median = 2.011. All USDA commodities remain below the original paper's global high-intensity threshold, driven by crops such as cocoa, coffee, and palm oil.

USDA Extension: Empirical Strategy

The specification mirrors the replication event study, adding a local market dimension:

$$\text{CashReturn}_{i,\ell,t} = \beta \left(\text{PostKunming}_t \times \text{HighIntensity}_i \right) + \alpha_{i,\ell} + \delta_t + \varepsilon_{i,\ell,t}$$

- i : commodity; ℓ : local U.S. market; t : month.
- $\text{PostKunming}_t = 1$ from **November 2021 to March 2022** (5-month window to capture gradual pass-through in local cash markets, which adjust more slowly than international spot prices).
- HighIntensity_i : above-median biodiversity intensity within the USDA sample.
- Fixed effects: commodity $\times$ local market ($\alpha_{i,\ell}$) absorbs all time-invariant heterogeneity at the unit level; month FE (δ_t) removes common shocks.
- SE clustered at the commodity $\times$ local market unit.

A continuous version replaces the indicator with the raw intensity value. A third specification uses **basis changes** (cash minus futures) as the outcome, providing evidence on the cash-futures link.

USDA Extension: Main Results

Table: USDA AMS Kunming event study — coefficient on Post $\times$ Intensity

Specification	Coef.	SE	Obs.	Adj. R^2
<i>Main grain sample, 6 commodities</i>				
Continuous intensity	−0.0043***	0.0007	4,388	0.625
High intensity dummy	−0.0243***	0.0021	4,388	0.630
<i>Expanded cash sample, 13 commodities</i>				
Continuous intensity	−0.0012	0.0014	4,905	0.464
High intensity dummy	−0.0146***	0.0043	4,905	0.466
<i>Basis changes, main grain</i>				
Continuous intensity	−0.0004	0.0034	4,672	0.308
High intensity dummy	−0.0473***	0.0096	4,672	0.311

FE: commodity $\times$ local market and month. Clustered SE. *** $p < 0.01$.

The main grain result is robust and economically meaningful. The expanded sample is noisier, but the high-intensity dummy remains significant; basis results suggest transmission into cash–futures spreads.

Robustness: Pre-Kunming Placebo Windows

The same regression is run on false pre-event windows entirely within the pre-Kunming period. If the main result were spurious, we would expect comparable negative effects at arbitrary pre-event dates.

Table: Pre-Kunming placebo windows — main grain, high intensity dummy

Placebo window	Coef.	SE	<i>p</i> -value	Obs.
Mar–May 2021	+0.0149	0.0038	0.0001	3,776
Apr–Jun 2021	+0.0423	0.0033	0.0000	3,776
May–Jul 2021	+0.0179	0.0024	0.0000	3,776

All placebo coefficients are *positive* — the opposite sign to the Kunming result of -0.0243 . High-intensity commodities were actually outperforming in the months before the event.

This sign reversal strengthens the causal interpretation: the negative repricing is specific to the post-Kunming period and not a continuation of a pre-existing trend.

Robustness: Leave-One-Commodity-Out

The main grain sample contains only six commodities. To verify that no single commodity drives the result:

Table: Leave-one-commodity-out — main grain, high-intensity dummy

Dropped commodity	Coef.	SE	<i>p</i> -value
Barley	−0.0240***	0.0021	0.0000
Corn	−0.0413***	0.0032	0.0000
Oats	−0.0254***	0.0019	0.0000
Sorghum	−0.0253***	0.0022	0.0000
Soybeans	−0.0164***	0.0041	0.0001
Wheat	−0.0149***	0.0018	0.0000

*** $p < 0.01$.

The coefficient is negative and highly significant regardless of which commodity is excluded. The largest attenuation occurs when corn is dropped — the highest-intensity and most widely traded commodity in the sample — yet even in that case the effect remains −2.4 pp and strongly significant.

Conclusion

What the replication confirms

- Biodiversity intensity earns a positive and significant premium in commodity spot returns.
- The premium emerges after 2015 and becomes large after global biodiversity policy milestones.
- The Kunming Declaration triggers a sharp negative repricing for high-intensity commodities, with no pre-event trend.

What the extension adds

- The repricing also appears in U.S. local cash markets, providing **external validation**.
- Results are strongest in the homogeneous main grain sample and robust to placebo and leave-one-out checks.
- Basis results suggest transmission from cash markets into cash–futures spreads.

Take-away

Biodiversity-transition risk is priced in commodity markets and responds to policy events, even at local physical delivery level.